

SIGNED CONE CUTS AS LORENTZ ORBITS: EIGEN-COORDINATES AND POWER MAPS

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ABSTRACT. The cutting-plane theorem of [1] sends a factor pair (x, y) to

$$X = \frac{x-y}{2}, \quad Y = \sqrt{xy}, \quad T = \frac{x+y}{2},$$

so that $X^2 + Y^2 = T^2$, and classifies the section cut by a line $ax + by = c$ by the sign of ab . We place that result inside the continuous Lorentz symmetry of the cone. After adjoining the reflected sheet $Y < 0$, every nondegenerate ellipse or hyperbola cut is a full orbit of the explicit generator

$$G_{a,b} = (a+b)L + (a-b)B_Y \in \mathfrak{so}(2,1),$$

whose characteristic polynomial is $\lambda(\lambda^2 + 4ab)$. Thus the conic classification, the causal type of the cutting-plane normal, and the eigenvalue type of the Lorentz generator are the same invariant. We diagonalize the elliptic and hyperbolic flows in complex and split eigen-coordinates, respectively. Integer powers of those coordinates define semiconjugate power maps once a particular equation (a, b, c) is chosen; the construction is not canonical under arbitrary rescaling of that equation. In the parabolic boundary case the diagonal eigen-coordinate construction collapses because the generator is nilpotent, although coordinatewise factor powers $(x, y) \mapsto (x^n, y^n)$ still exist. The same rapidity $s = \frac{1}{2} \log(x/y)$ used here is the continuous Lorentz coordinate that later supports the discriminant-12 arithmetic return.

1. FROM THE FACTOR SHEET TO THE SIGNED CONE

For positive factor coordinates define

$$(1) \quad X = \frac{x-y}{2}, \quad Y = \sqrt{xy}, \quad T = \frac{x+y}{2}.$$

Then

$$(2) \quad X^2 + Y^2 = T^2, \quad x = T + X, \quad y = T - X.$$

Paper A [1] uses the positive-factor sheet $Y \geq 0$. The Lorentz group acts more naturally after adjoining its reflection across $Y = 0$.

Definition 1.1 (Signed future cone). *Let*

$$\mathcal{C}^+ = \{(X, Y, T) \in \mathbb{R}^3 : T > 0, X^2 + Y^2 = T^2\},$$

with Y allowed to have either sign. The Paper-A factor image is the half-sheet

$$\mathcal{C}_{\text{fac}}^+ = \mathcal{C}^+ \cap \{Y \geq 0\}.$$

A line $ax + by = c$ becomes, by (2),

$$(3) \quad (a-b)X + (a+b)T = c.$$

The corrected cutting-plane classification is: $ab > 0$ gives an ellipse, $ab = 0$ a parabola, $ab < 0$ with $c \neq 0$ a nondegenerate hyperbola, while $ab < 0, c = 0$ is the degenerate generator case [1].

2. THE CONE SYMMETRY ALGEBRA

Let

$$\eta = \text{diag}(1, 1, -1),$$

so the cone is the null set of the quadratic form $X^2 + Y^2 - T^2$. Define

$$L = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B_X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad B_Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Each satisfies

$$G^T \eta + \eta G = 0,$$

and therefore belongs to $\mathfrak{so}(2, 1)$. Their brackets are

$$[L, B_X] = B_Y, \quad [L, B_Y] = -B_X, \quad [B_X, B_Y] = -L.$$

The three elementary flows have a direct factor interpretation. The L -flow fixes T , hence fixes $x + y$; the B_X -flow fixes Y , hence fixes xy ; and the B_Y -flow fixes X , hence fixes $x - y$. In particular, if

$$s = \frac{1}{2} \log \frac{x}{y},$$

then the B_X boost acts by translation of rapidity,

$$x \mapsto e^\phi x, \quad y \mapsto e^{-\phi} y, \quad s \mapsto s + \phi.$$

3. THE GENERATOR ATTACHED TO A CUTTING DIRECTION

Definition 3.1. For real a, b , not both zero, set

$$(4) \quad G_{a,b} := (a + b)L + (a - b)B_Y.$$

Writing

$$\alpha = a + b, \quad \gamma = a - b,$$

we have

$$G_{a,b} = \begin{pmatrix} 0 & -\alpha & 0 \\ \alpha & 0 & \gamma \\ 0 & \gamma & 0 \end{pmatrix}.$$

Proposition 3.2 (Invariant cutting plane). *The vector*

$$m = (a - b, 0, -(a + b)) = (\gamma, 0, -\alpha)$$

lies in the kernel of $G_{a,b}$. Consequently the Lorentz functional

$$\langle m, (X, Y, T) \rangle_\eta = (a - b)X + (a + b)T$$

is constant along every $G_{a,b}$ flow line, so every orbit stays in one plane (3).

Proof. For $m = (m_1, 0, m_3)$ the equation $Gm = 0$ is exactly

$$-\alpha \cdot 0 = 0, \quad \alpha m_1 + \gamma m_3 = 0, \quad \gamma \cdot 0 = 0.$$

Taking $(m_1, m_3) = (\gamma, -\alpha)$ satisfies the middle equation. Since $G^T \eta + \eta G = 0$,

$$\frac{d}{dt} \langle m, e^{tG} p \rangle_\eta = \langle m, G e^{tG} p \rangle_\eta = -\langle Gm, e^{tG} p \rangle_\eta = 0.$$

□

Proposition 3.3 (One invariant, three classifications). *The characteristic polynomial of $G_{a,b}$ is*

$$(5) \quad \boxed{\chi_{G_{a,b}}(\lambda) = \lambda(\lambda^2 + 4ab).}$$

The plane normal

$$n = (a - b, 0, a + b)$$

has Lorentz norm

$$(6) \quad \boxed{\langle n, n \rangle_\eta = -4ab.}$$

Hence:

- (i) $ab > 0$: n is timelike and the nonzero eigenvalues are $\pm 2i\sqrt{ab}$;
- (ii) $ab < 0$: n is spacelike and the nonzero eigenvalues are $\pm 2\sqrt{-ab}$;
- (iii) $ab = 0$: n is null and $G_{a,b}$ is nilpotent.

Thus Paper A's ellipse/parabola/hyperbola trichotomy is exactly the elliptic/parabolic/hyperbolic trichotomy of the associated Lorentz generator.

Proof. A determinant expansion gives (5). Directly,

$$(a - b)^2 - (a + b)^2 = -4ab,$$

which gives (6); the three cases follow. □

Common rescaling does not change the cutting direction:

$$(a, b) \mapsto (\kappa a, \kappa b) \implies G_{\kappa a, \kappa b} = \kappa G_{a, b}.$$

Thus the underlying one-parameter subgroup is naturally attached to the projective direction $[a : b]$, while the matrix generator is fixed only up to time rescaling.

4. SIGNED CUTS ARE COMPLETE LORENTZ ORBITS

Theorem 4.1 (Orbit theorem). *Assume $ab \neq 0$ and that the plane (3) meets the future cone in a nondegenerate section. On the signed future cone $\mathcal{C}^+$, each connected nondegenerate section is a full orbit of the one-parameter subgroup $e^{tG_{a,b}}$. The multiplication-table trace of Paper A is the portion of that orbit lying on $Y \geq 0$.*

Proof. Proposition 3.2 shows that every orbit is contained in the plane section. If $ab > 0$, Proposition 3.3 gives an elliptic one-parameter subgroup; after a Lorentz change of frame sending the timelike plane normal to the time axis, the section is a Euclidean circle and the subgroup is an ordinary rotation, hence transitive on it. If $ab < 0$, the normal is spacelike; after a Lorentz change of frame sending it to a spatial axis, the section is the future branch of a hyperbola in a timelike two-plane and the subgroup is an ordinary boost, again transitive on that connected branch. Returning by the inverse Lorentz change of frame proves the claim. Restricting to $Y \geq 0$ gives the Paper-A factor trace. □

Remark. The signed-sheet qualification is essential. For $a = b$, $G_{a,a} = 2aL$ rotates the full signed circle, whereas the factor map $Y = +\sqrt{xy}$ displays only its upper semicircle. Likewise, $a = -b$ gives the full signed B_Y hyperbola, while the factor sheet retains only $Y \geq 0$.

5. EIGEN-COORDINATES

5.1. **Elliptic case.** Assume $ab > 0$ and define

$$(7) \quad \zeta_{a,b} = (a+b)X + (a-b)T + 2i\sqrt{ab}Y.$$

Proposition 5.1 (Complex eigen-coordinate). *Along the $G_{a,b}$ flow,*

$$(8) \quad \zeta(t) = e^{2i\sqrt{ab}t}\zeta(0).$$

On the plane (3),

$$(9) \quad \boxed{|\zeta_{a,b}|^2 = c^2, \quad |\zeta_{a,b}| = |c|}.$$

Proof. The row vector

$$w^\top = (a+b, \ 2i\sqrt{ab}, \ a-b)$$

satisfies

$$w^\top G_{a,b} = 2i\sqrt{ab} w^\top,$$

which gives (8). For the norm, write

$$P = (a+b)X + (a-b)T, \quad Q = (a-b)X + (a+b)T = c.$$

A direct expansion gives

$$P^2 - Q^2 = 4ab(X^2 - T^2) = -4abY^2,$$

so

$$|\zeta|^2 = P^2 + 4abY^2 = Q^2 = c^2.$$

□

Thus the ellipse becomes a circle in its own natural eigen-coordinate.

5.2. **Hyperbolic case.** Assume $ab < 0$, put

$$r = \sqrt{-ab},$$

and define

$$(10) \quad \eta_\pm = (a+b)X + (a-b)T \pm 2rY.$$

Proposition 5.2 (Split eigen-coordinates). *Along the flow,*

$$\eta_+(t) = e^{2rt}\eta_+(0), \quad \eta_-(t) = e^{-2rt}\eta_-(0),$$

and on the cutting plane,

$$(11) \quad \boxed{\eta_+\eta_- = c^2}.$$

Proof. The corresponding row vectors are left eigenvectors of $G_{a,b}$ with eigenvalues $\pm 2r$. With the same P, Q as above and $r^2 = -ab$,

$$\eta_+\eta_- = P^2 - 4r^2Y^2 = P^2 + 4abY^2 = Q^2 = c^2.$$

□

The two cases therefore have parallel normal forms:

$$|\zeta|^2 = c^2 \quad \text{or} \quad \eta_+\eta_- = c^2.$$

6. POWER MAPS ATTACHED TO A CHOSEN EQUATION

A geometric line is unchanged by

$$(a, b, c) \mapsto (\kappa a, \kappa b, \kappa c), \quad \kappa \neq 0.$$

The power construction below is therefore attached to a *chosen equation* (a, b, c) , not canonically to the unnormalized geometric line. This distinction is necessary: the eigen-coordinates scale linearly with the coefficients, while their n th powers scale by κ^n .

Throughout this section let

$$n \in \mathbb{Z}_{\geq 1}.$$

6.1. Elliptic power map. Assume $ab > 0$. Given a point on the signed section, define

$$\zeta_n := \zeta_{a,b}^n = P_n + iR_n.$$

Define (X_n, Y_n, T_n) by

$$(12) \quad (a+b)X_n + (a-b)T_n = P_n,$$

$$(13) \quad (a-b)X_n + (a+b)T_n = c^n,$$

$$(14) \quad 2\sqrt{ab}Y_n = R_n.$$

The 2×2 coefficient matrix in (12)–(13) has determinant $4ab \neq 0$, so the reconstruction is unique.

Proposition 6.1 (Elliptic power map). *The reconstructed point satisfies*

$$X_n^2 + Y_n^2 = T_n^2$$

and lies on the chosen target plane

$$(a-b)X_n + (a+b)T_n = c^n,$$

equivalently $ax_n + by_n = c^n$ for $x_n = T_n + X_n$, $y_n = T_n - X_n$.

Proof. Because $|\zeta_n|^2 = |\zeta|^{2n} = c^{2n}$,

$$P_n^2 + R_n^2 = c^{2n}.$$

Let

$$Q_n = (a-b)X_n + (a+b)T_n = c^n.$$

The identity used in Proposition 5.1 reverses:

$$P_n^2 - Q_n^2 = 4ab(X_n^2 - T_n^2).$$

Since $R_n = 2\sqrt{ab}Y_n$ and $P_n^2 + R_n^2 = Q_n^2$, we obtain

$$4ab(X_n^2 - T_n^2) = -4abY_n^2,$$

hence the cone equation. □

6.2. Hyperbolic power map. Assume $ab < 0$. Define

$$\eta_{+,n} := \eta_+^n, \quad \eta_{-,n} := \eta_-^n,$$

then set

$$(15) \quad P_n = \frac{\eta_{+,n} + \eta_{-,n}}{2}, \quad Y_n = \frac{\eta_{+,n} - \eta_{-,n}}{4\sqrt{-ab}}.$$

Reconstruct X_n, T_n from

$$(16) \quad (a+b)X_n + (a-b)T_n = P_n,$$

$$(17) \quad (a-b)X_n + (a+b)T_n = c^n.$$

The coefficient determinant is again $4ab \neq 0$.

Proposition 6.2 (Hyperbolic power map). *The reconstructed point lies on the cone and on the chosen target plane with right-hand side c^n .*

Proof. From (11),

$$\eta_{+,n}\eta_{-,n} = c^{2n}.$$

Equation (15) gives

$$P_n^2 - 4(-ab)Y_n^2 = c^{2n}.$$

Writing $Q_n = c^n$, this is

$$P_n^2 + 4abY_n^2 = Q_n^2.$$

Using again

$$P_n^2 - Q_n^2 = 4ab(X_n^2 - T_n^2)$$

gives $X_n^2 + Y_n^2 = T_n^2$. □

6.3. Semiconjugacy with the Lorentz flow.

Theorem 6.3 (Power-flow semiconjugacy). *For either nonparabolic eigen-coordinate construction,*

$$(18) \quad \boxed{\Psi_n \circ \text{flow}_t = \text{flow}_{nt} \circ \Psi_n.}$$

Proof. In an eigen-coordinate z with flow $z(t) = e^{\omega t} z_0$,

$$\Psi_n(z(t)) = (e^{\omega t} z_0)^n = e^{\omega n t} z_0^n.$$

The right-hand side is exactly the time- nt flow starting from the output point $\Psi_n(z_0) = z_0^n$. The split case is the same calculation on the pair (η_+, η_-) with opposite exponents. □

Remark (Normalization guardrail). If the same geometric line is rewritten as $(\kappa a)x + (\kappa b)y = \kappa c$, then the chosen eigen-coordinate is rescaled before taking its n th power. For $n > 1$ this changes the target geometric level. Thus (18) is exact for a fixed chosen equation, but the map is not invariant under arbitrary rescaling of that equation.

7. THE PARABOLIC BOUNDARY

Suppose $ab = 0$. Then Proposition 3.3 gives only the zero eigenvalue and $G_{a,b}$ is nilpotent. The complex/split diagonal eigen-coordinate constructions above therefore collapse.

Proposition 7.1 (What degenerates at a row or column). *For $ab = 0$, there is no nonzero diagonal eigen-coordinate of the elliptic/hyperbolic type used in Sections 5–6. Consequently the preceding eigen-coordinate power-map construction does not extend by continuity as a diagonalization argument.*

This does *not* mean that no power map exists. The elementary coordinatewise operation

$$(19) \quad \boxed{P_n(x, y) = (x^n, y^n)}$$

sends a row $x = c$ to $x = c^n$, and a column $y = c$ to $y = c^n$. Its cone coordinates are

$$X_n = \frac{x^n - y^n}{2}, \quad Y_n = (xy)^{n/2}, \quad T_n = \frac{x^n + y^n}{2},$$

which satisfy $X_n^2 + Y_n^2 = T_n^2$ identically.

There are also two distinct one-parameter notions in the parabolic case. Literal translation $y \mapsto y + d$ preserves a row in factor coordinates, whereas $e^{tG_{a,0}}$ is the nilpotent Lorentz null-rotation flow on the signed cone. They should not be identified without an explicit parameter correspondence.

8. CONSTANT PRODUCT AND RAPIDITY

The coordinatewise power map (19) is particularly transparent on a constant-product curve:

$$xy = N \implies x^n y^n = N^n.$$

In rapidity,

$$s = \frac{1}{2} \log \frac{x}{y},$$

we obtain

$$(20) \quad \boxed{s \mapsto ns.}$$

Thus coordinatewise powers dilate Lorentz rapidity while raising the product level. This is structurally different from the chosen-equation eigen-coordinate powers of Sections 6, even though both are naturally expressed in coordinates adapted to one-parameter Lorentz actions.

The rapidity variable is also the one used by the area companion [2], where it compactifies into circle angle through

$$\cos \theta = \operatorname{sech} s, \quad d\theta = \operatorname{sech} s \, ds.$$

This agreement is an exact coordinate bridge across the continuous Cone framework.

9. RELATION TO THE DISCRIMINANT-12 RETURN

The present paper is continuous: it identifies the real Lorentz flows naturally attached to Paper-A cutting directions. A separate arithmetic construction [3] selects a distinguished integral return with discriminant 12 and real boost factor

$$\lambda = 2 + \sqrt{3}.$$

In factor light-cone coordinates that return acts as

$$x \mapsto \lambda x, \quad y \mapsto \lambda^{-1} y,$$

so its rapidity action is

$$s \mapsto s + \log \lambda.$$

This is a discrete arithmetic element represented inside the same real Lorentz geometry; it is not literally the generic continuous generator $G_{a,b}$ of a cutting plane.

10. CONCLUSION

The continuous symmetry behind Paper A is now explicit. The cutting-plane normal has Lorentz norm $-4ab$, the corresponding generator has characteristic polynomial $\lambda(\lambda^2 + 4ab)$, and the signed nondegenerate section is a full orbit of that generator. Complex and split eigen-coordinates reduce the elliptic and hyperbolic cases to $|\zeta|^2 = c^2$ and $\eta_+ \eta_- = c^2$. Integer powers give exact semi-conjugate maps once a particular equation (a, b, c) is fixed, while the parabolic boundary marks the failure of diagonalization rather than the failure of all power operations. Together with the common rapidity variable, these results make the present paper the continuous Lorentz-symmetry companion to the elementary geometry of Paper A and the later arithmetic return construction.

REFERENCES

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- [3] J. Ebert, *The Discriminant-12 Return: From a Centered Cayley Selection to a Cyclotomic and Finite Arithmetic Structure*, 2026.